

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – II - Binary Codes

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3. Binary Codes

Video Link - <https://youtu.be/uN4oWD> CC1g

- When numbers, alphabets or words are represented by a specific group of symbols, we can say that they are encoded.
- The group of symbols used to encode them are called codes. The digital data is represented, stored and transmitted as groups of binary digits (bits).
- The group of bits, also known as binary code, represent both numbers and letters of the alphabets as well as many special characters and control functions.

Classification of Binary Codes

(a) Weighted Codes

(i) Binary

(ii) BCD - Binary Coded Decimal

8421

2421

3321

4221

5211

5311

5421

6311

7421

74 $\bar{2}\bar{1}$

84 $\bar{2}\bar{1}$

(b) Non - Weighted Codes

(i) Excess - 3

(ii) Gray Codes

(c) Reflective Codes

(i) 2421

(ii) 5211

(iii) Excess - 3

(d) Sequential Codes

(i) 8421

(ii) Excess - 3

(e) Alphabetic Codes

(i) ASCII

(ii) EBCDIC

(iii) Hollerith Code

(f) Error detecting and Correcting Codes

(i) Parity

(ii) Hamming

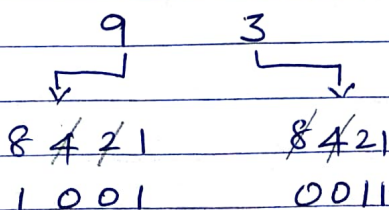
Details of Binary Codes

(a) Weighted codes

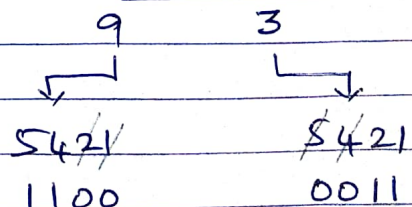
→ In weighted Codes, each digit position of the number represents a specific weight

→ In weighted binary code each bit has a weight 8, 4, 2, ~~or~~^{or} 1 and each decimal digit is represented by a group of four bits

8421 Code

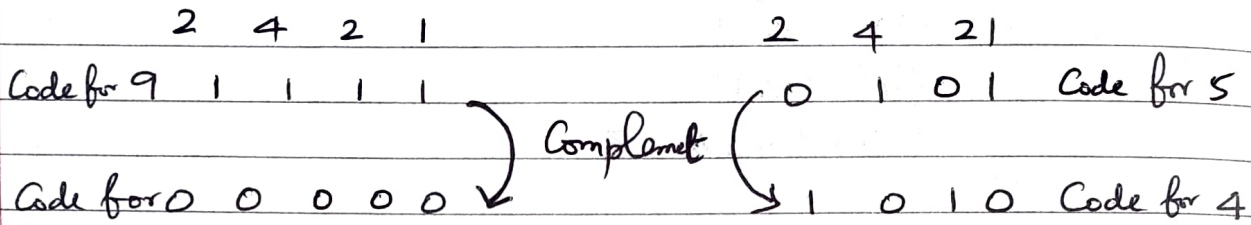


5421 Code



(b) Reflective Codes - 2421, 5211 and Excess-3

→ A Code is said to be reflective when Code for 9 is the Complement for 0, the Code for 8 is Complement for 1, 7 for 2, 6 for 3, 5 for 4.



→ Like 2421, Code 5211 and excess-3 are also reflective.

→ The Code 8421 is not reflective

→ Reflectivity is desirable in a code when the nine's Complement must be found, such as in nine's Complement Subtraction.

(i) 2421

Decimal	2 4 2 1
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	1 0 1 1
6	1 1 0 0
7	1 1 0 1
8	1 1 1 0
9	1 1 1 1

(ii) 5211

Decimal	5	2	1	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	1	0
4	0	1	1	1
5	1	0	0	0
6	1	0	0	1
7	1	1	0	0
8	1	1	1	0
9	1	1	1	1

(iii) Excess-3 Code

Decimal	Excess-3 Code
	8 4 2 1
0(3)	0 0 1 1
1(4)	0 1 0 0
2(5)	0 1 0 1
3(6)	0 1 1 0
4(7)	0 1 1 1
5(8)	1 0 0 0
6(9)	1 0 0 1
7(10)	1 0 1 0
8(11)	1 0 1 1
9(12)	1 1 0 0

(i) Sequential Codes

Video Link - <https://youtu.be/HhkpBR2z244>

→ In sequential code each succeeding code is one binary number greater than its preceding code

→ This greatly aids mathematical manipulation of data

→ The 8421 and Encus are sequential, whereas the 2421 and 5211 codes are not sequential

(i) 8421

Decimal	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

(ii) Encus-3 Code

Decimal	Encus-3
	8 4 2 1
0(3)	0 0 1 1
1(4)	0 1 0 0
2(5)	0 1 0 1
3(6)	0 1 1 0
4(7)	0 1 1 1
5(8)	1 0 0 0
6(9)	1 0 0 1
7(10)	1 0 1 0
8(11)	1 0 1 1
9(12)	1 1 0 0

(d) Alphanumeric Codes

- The Codes which consists of both numbers and alphabetic characters are called alphanumeric codes
- The most commonly used alphanumeric codes are
 - (i) ASCII - American Standard Code for Information Interchange
 - (ii) EBCDIC - Extended Binary Coded Decimal Interchange Code

(iii) Hollerith Code

(e) Error detecting and Correcting Codes

- When the digital information in the binary form is transmitted from one circuit or system to another circuit or system an error may occur. This means a signal corresponding to 0 may change to 1 or vice versa due to presence of noise
- To maintain the data integrity between transmitter and receiver, extra bit or more than one bit are added in the data
- These extra bits allow the detection and sometimes correction of errors in the data
- The data along with the extra bit/bits forms the code. Codes which allow only error detection are called error detecting codes and codes which allow error detection and correction are called error detecting and correcting codes.

Binary Codes

Video Link - <https://youtu.be/nuGscyx7PU8>

(a) Binary Coded Decimal - BCD

(i) 8421

Decimal	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

(ii) 2421

Decimal	2	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	0	1
8	1	1	1	0
9	1	1	1	1

(iii) 3321

Decimal	3	3	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	0	1
8	1	1	1	0
9	1	1	1	1

(iv) 4221

Decimal	4	2	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	1	1
6	1	1	0	0
7	1	1	0	1
8	1	1	1	0
9	1	1	1	1

(v) 5211

<u>Decimal</u>	<u>5</u>	<u>2</u>	<u>1</u>	<u>1</u>
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	1	0
4	0	1	1	1
5	1	0	0	0
6	1	0	0	1
7	1	1	0	0
8	1	1	1	0
9	1	1	1	1

(vi) 5311

<u>Decimal</u>	<u>5</u>	<u>3</u>	<u>1</u>	<u>1</u>
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	0	0
4	0	1	0	1
5	1	0	0	0
6	1	0	0	1
7	1	0	1	1
8	1	1	0	0
9	1	1	0	1

(vii) 5421

<u>Decimal</u>	<u>5</u>	<u>4</u>	<u>2</u>	<u>1</u>
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	1	0	0	0
6	1	0	0	1
7	1	0	1	0
8	1	0	1	1
9	1	1	0	0

(viii) 6311

<u>Decimal</u>	<u>6</u>	<u>3</u>	<u>1</u>	<u>1</u>
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	0	0
4	0	1	0	1
5	0	1	1	1
6	1	0	0	0
7	1	0	0	1
8	1	0	1	1
9	1	1	0	0

(ix) 7421

<u>Decimal</u>	<u>7</u>	<u>4</u>	<u>2</u>	<u>1</u>
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	1	0	0	0
8	1	0	0	1
9	1	0	1	0

(x) 74 $\bar{2}\bar{1}$

<u>Decimal</u>	<u>7</u>	<u>4</u>	<u>$\bar{2}$</u>	<u>$\bar{1}$</u>
0	0	0	0	0
1	0	1	1	1
2	0	1	1	0
3	0	1	0	1
4	0	1	0	0
5	1	0	1	0
6	1	0	0	1
7	1	0	0	0
8	1	1	1	1
9	1	1	1	0

(xi) 84 $\bar{2}\bar{1}$

<u>Decimal</u>	<u>8</u>	<u>4</u>	<u>$\bar{2}$</u>	<u>$\bar{1}$</u>
0	0	0	0	0
1	0	1	1	1
2	0	1	1	0
3	0	1	0	1
4	0	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	1	1	1	1

(xii) Excess-3 Codes

<u>Decimal</u>	<u>Excess-3</u>			
	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>
0(3)	0	0	1	1
1(4)	0	1	0	0
2(5)	0	1	0	1
3(6)	0	1	1	0
4(7)	0	1	1	1
5(8)	1	0	0	0
6(9)	1	0	0	1
7(10)	1	0	1	0
8(11)	1	0	1	1
9(12)	1	1	0	0

Gray Code

Video Link - <https://youtu.be/QgHHkUOfFYI>

→ Gray Code is a non-weighted Code and is a Special Case of Unit-distance Code

→ In Unit-distance code, bit patterns for two Consecutive numbers differ in only one bit position. These Codes are also called Cyclic Codes

Decimal	Building Gray Bits				Gray Code			
0				0	0	0	0	0
1				1	0	0	0	1
2			1	1	0	0	1	1
3			1	0	0	0	1	0
4		1	1	0	0	1	1	0
5		1	1	1	0	1	1	1
6		1	0	1	0	1	0	1
7		1	0	0	0	1	0	0
8	1	1	0	0	1	1	0	0
9	1	1	0	1	1	1	0	1
10	1	1	1	1	1	1	1	1
11	1	1	1	0	1	1	1	0
12	1	0	1	0	1	0	1	0
13	1	0	1	1	1	0	1	1
14	1	0	0	1	1	0	0	1
15	1	0	0	0	1	0	0	0

Alphanumeric Codes

Video Link - <https://youtu.be/F0LSWYN5N1s>

→ The codes which consists of both numbers and alphabetic characters are called alphanumeric codes

→ The most commonly used alphanumeric codes are

(i) ASCII - American Standard Code for Information Interchange

(ii) EBCDIC - Extended Binary Coded Decimal Interchange Code

(iii) Hollerith Code

→ These codes consist of symbols to represent

- * 26 alphabets with capital and small letters
- * Numbers from 0 to 9
- * Punctuation marks and other symbols

(a) ASCII

→ ASCII code is a seven bit code in which the decimal digits are represented by the BCD

→ Since it is a 7-bit code, it represents
 $2^7 = 128$ symbols

→ The letters of the alphabets and other symbols and instructions are represented by other code combinations as shown in table

For example, ASCII code for

$$N = (4E)_{16} = \begin{array}{cc} 1001 & 110 \\ \text{MSB} & \text{LSB} \end{array}$$

LSBs				MSBs				
	000(0)	001(1)	010(2)	011(3)	100(4)	101(5)	110(6)	111(7)
0000 (0)	NUL	DLE	SP	0	@	P	'	p
0001 (1)								
0010 (2)	SOH	DC ₁	!	1	A	Q	a	q
0010 (2)	STX	DC ₂	"	2	B	R	b	r
0011 (3)	ETX	DC ₃	#	3	C	S	c	s
0100 (4)	EBT	DC ₄	\$	4	D	T	d	t
0101 (5)	ENQ	NAK	%	5	E	U	e	u
0110 (6)	ACK	SYN	&	6	F	V	f	v
0111 (7)	BEL	ETB	'	7	G	W	g	w
1000 (8)	BS	CAN	(	8	H	X	h	x
1001 (9)	HT	EM	)	9	I	Y	i	y
1010 (A)	LF	SUB	*	:	J	Z	j	z
1011 (B)	VT	ESC	+	;	K	[	k	{
1100 (C)	FF	FS	,	<	L	\	l	
1101 (D)	CR	GS	-	=	M	]	m	}
1110 (E)	SO	RS	.	>	N	^	n	-
1111 (F)	SI	US	/	?	O	_	o	DEL

Definition of Control Abbreviations

ACK - Acknowledge	FS - Form Separator
BEL - Bell	GS - Group Separator
BS - Backspace	HT - Horizontal tab
CAN - Cancel	LF - Line feed
CR - Carriage return	NAK - Negative acknowledge
DC ₁ -DC ₄ Direct Control	NUL - Null
DEL - Delete idle	RS - Record separator
DLE - Data Link escape	SI - Shift in
EM - End of medium	SO - Shift out
ENQ - Enquiry	SOH - Start of heading
EOT - End of Transmission	STX - Start text
ESC - Escape	SUB - Subtitle Substitute
ETB - End of Transmission block	SYN - Synchronous idle
ETX - End Text	US - Unit Separator
FF - Form feed	VT - Vertical tab

(b) EBCDIC

Video Link - https://youtu.be/Ab_0RW1lcQ

→ EBCDIC is a Coding System used to represent characters, letters, numerals, punctuation marks, and other symbols in computerized text

→ EBCDIC is a 8 bit code

→ $2^8 = 256$ 256 different characters are available

→ EBCDIC mainly used on IBM mainframe and IBM mid-range Computer operating systems.

Character	EBCDIC Bit Configuration	Character	EBCDIC Bit Configuration
A	1100 0001	V	1110 0101
B	1100 0010	W	1110 0110
C	1100 0011	X	1110 0111
D	1100 0100	Y	1110 1000
E	1100 0101	Z	1110 1001
F	1100 0110	[	1111 0000
G	1100 0111	]	1111 0001
H	1100 1000	^	1111 0010
I	1100 1001	_	1111 0011
J	1101 0001	`	1111 0100
K	1101 0010	a	1111 0101
L	1101 0011	b	1111 0110
M	1101 0100	c	1111 0111
N	1101 0101	d	1111 1000
O	1101 0110	e	1111 1001
P	1101 0111	f	
Q	1101 1000		
R	1101 1001		
S	1110 0010		
T	1110 0011		
U	1110 0100		

(c) Hollerith Code

A System for coding data into punched cards, in which each horizontal row is assigned a different value, and letters, numbers, or special characters are encoded as combinations of these values in a vertical column.

Video Link - <https://youtu.be/Ghumk7suKIk>

Error Detecting and Correcting Codes Parity Bit

- A parity bit is used for the purpose of detecting errors during transmission of binary information.
- A parity bit is an extra bit included with a binary message to make the number of 1's either odd or even.
- The message, including the parity bit is transmitted and then checked at the receiving end for errors.
- An error is detected if the checked parity does not correspond with one transmitted.
- The circuit that generates the parity bit in the transmitter is called a parity generator and the circuit that checks the parity in the receiver is called a parity checker.
- In even parity the added parity bit will make the total number of 1's an even amount.
- In odd parity the added parity bit will make the total number of 1's an odd amount.

Message with even and odd parity

3 bit Message			Message with odd parity		Message with Even parity	
A	B	C	Message	Parity	Message	Parity
0	0	0	000	1	000	0
0	0	1	001	0	001	1
0	1	0	010	0	010	1
0	1	1	011	1	011	0
1	0	0	100	0	100	1
1	0	1	101	1	101	0
1	1	0	110	1	110	0
1	1	1	111	0	111	1

Example 1

In an even parity scheme, which of the following words contain an error

(a) 10101010

No. of 1's is 4 (ie) Even So No error

(b) 11110110

No. of 1's is 6 (ie) Even So No error

(c) 10111001

No. of 1's is 5 (ie) Odd, So this word has an error

Example 2

In an odd parity scheme, which of the following words contain an error

(a) 10110111

No. of 1's is 6 (ie) Even, Has an error

(b) 10011010

No. of 1's is 4 (ie) Even, Has an error

(c) 11101010

No. of 1's is 5 (ie) Odd, Has no error

Hamming Code

Video Link - <https://youtu.be/AZRY-fAD5Mk>

- Hamming Code not only provides the detection of a bit error, but also identifies which bit is in error so that it can be corrected
- Thus Hamming Code is called error detecting and correcting code
- The code uses a number of parity bits located at certain positions in the code group

Number of Parity bits

Number of parity bits depend on the number of information bits

If the number of information bit is designed x , then the number of parity bits, P is given by

$$2^P \geq x + P + 1$$

Let $x = 4$

$$\text{Take } P = 2 \Rightarrow 2^2 \geq 4 + 2 + 1$$

$$4 \geq 7 \text{ is not satisfied}$$

$$\text{Take } P = 3 \Rightarrow 2^3 \geq 4 + 3 + 1$$

$$8 \geq 8 \text{ is satisfied}$$

Locations of the Parity Bits in the Code

For above example, we have four information bits and three parity bits

So the code is of seven bits

bit 7, bit 6, bit 5, bit 4, bit 3, bit 2, bit 1

The parity bits are located in the position that are numbered corresponding to ascending powers of two 2^0 2^1 2^2 2^3 ... (i.e) 1, 2, 4, 8....

Therefore, for 7 bit code, locations for parity bits and information bit are as follows

D7, D6, D5, P4, D3, P2, P1
~~2⁶~~ ~~2⁵~~ ~~2⁴~~ ~~2³~~ ~~2²~~ ~~2¹~~ ~~2⁰~~
 7 6 5 (4) 3 (2) (1)

Bit Designation D7 D6 D5 D4 D3 D2

Bit Designation	D7	D6	D5	P4	D3	P2	P1
Bit Location	7	6	5	4	3	2	1
Binary Location Number	111	110	101	100	011	010	001

Assignment of P1

From the above table, parity bit P1 has 1 for its right-most digit

This parity bit checks all bits locations, including itself, that have 1's in the same location

Therefore, parity bit P1 checks the bit locations, 1, 3, 5 and 7

Assigns P1 according to even or odd parity

Assignment of P2

From the above table, parity bit P2 has 1 for its middle bit.

This parity bit checks all bit locations, including itself, that have 1's in the middle bit.

Therefore, P2 checks 2, 3, 6 and 7

Assigns P2 according to even or odd parity

Assignment of P4

From the above table, parity bit P4 has 1 for its left-most bit.

This parity bit checks all bit locations, including itself, that have 1's in the left-most bit.

Therefore, parity bit P4 checks bit locations 4, 5, 6 and 7

Assigns P4 according to even or odd parity

Example 1

Video Link - <https://youtu.be/mbffTzlgKlg>

Encode the binary word 1011 into Seven bit even parity Hamming Code

Solution :

Step 1 - Find the number of parity bits required

Let $P = 3$ then

$$2^P \geq X + P + 1$$

$$X \text{ is } 4 \therefore 2^3 \geq 4 + 3 + 1$$

$$8 \geq 8 \text{ Satisfied}$$

$$\therefore \text{Total Code bits} = X + P$$

$$= 4 + 3$$

$$= 7$$

Step 2 - Construct a bit location table

Bit Designation	D7	D6	D5	P4	D3	P2	P1
Bit location	7	6	5	4	3	2	1
Binary location Number	111	110	101	100	011	010	001
Information bits	1	0	1		1		
Parity bits				0		0	1

$\therefore$ Seven bit Hamming Code is 1010101

Step 3 - Determine the parity bits

For P₁ - Bit locations 3, 5 and 7 have three 1's
therefore to have an even parity P₁ must be 1

For P₂ - Bit locations 3, 6 and 7 have two 1's and therefore
to have an even parity P₂ must be 0

For P₄ - Bit locations 5, 6 and 7 have two 1's and therefore
to have an even parity P₄ must be 0

Example 2

Video Link - <https://youtu.be/zM7r53umfnE>

Determine the Single Error-Correcting Code
for the information Code 10111 for odd parity

Solution

Step 1 - Find the number of parity bits required

Let $P = 3$ then

$$2^P \geq X + P + 1$$

$$X = 5 \quad \therefore 2^3 \geq 5 + 3 + 1$$

$$8 \geq 9 \quad \text{not Satisfied}$$

Let $P = 4$ then

$$2^P \geq X + P + 1$$

$$2^4 \geq 5 + 4 + 1$$

$$16 \geq 10 \quad \text{is Satisfied}$$

$$\therefore \text{Total Code bit} = X + P = 5 + 4 = 9$$

Big Designation	D ₉	P ₈	D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
Bit Location	9	8	7	6	5	4	3	2	1
Location Number	1001	1000	0111	0110	0101	0100	0011	0010	0001
Information bits	1		0	1	1		1		
Parity bits		0				1		1	0

$\therefore$ Nine bit Hamming Code is 1001 1111 0

For P1 - Bit locations 3, 5, 7 and 9 have three 1's and therefore to have odd parity P1 must be 0

For P2 - Bit locations 3, 6, 7 have two 1's and therefore to have an odd parity P2 must be 1

For P4 - Bit locations 5, 6 and 7 have two 1's and therefore to have an odd parity P4 must be 1

For P8 - Bit locations 8 and 9 have one 1's and therefore to have an odd parity P8 must be 0

Detecting and Correcting an Error

Example 1

Video Link - <https://youtu.be/asCdXuJ1zsM>

Assume that the even parity Hamming Code is example 0110 011 is transmitted and that 0100011 is received. The receiver does not know what is transmitted. Determine bit location where error has occurred using received code

Bit Designation	D7	D6	D5	P4	D3	P2	P1
Bit Location	7	6	5	4	3	2	1
Location Number	111	110	101	100	011	010	001
Received Code	0	1	0	0	0	1	1

For P1 - P1 checks locations 1, 3, 5, 7

There is one 1 in the group

$\therefore$ Parity check for even parity is wrong

$\therefore$ 1 (LSB)

For P2 - P2 checks locations 2, 3, 6, and 7

There are two 1's in the group

∴ Parity check for even parity is Correct $\rightarrow 0$

For P4 - P4 checks locations 4, 5, 6 and 7

There are one 1's in the group

∴ Parity check for even parity is wrong $\rightarrow 1$ (MSB)

The resultant word is 101.

This says that the bit in the number 5 location is in error. It is 0 and should be a 1.

Therefore, the Correct Code is 0110011, which agrees with the transmitted Code

Example 2

Video Link - <https://youtu.be/p0lop2n88rQ>

The Hamming Code 101101101 is received. Correct it if any errors. There are four parity bits and odd parity is used

Bit Designation	D9	P8	D7	D6	D5	P4	D3	P2	P1
	9	8	7	6	5	4	3	2	1
	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received Code	1	0	1	1	0	1	1	0	1

For P1 - P1 checks locations 1, 3, 5, 7 and 9

There are four 1's in the group

∴ Parity check for odd parity is wrong 1 (LSB)

For P2 - P2 checks locations 2, 3, 6 and 7

There are three 1's in the group

∴ Parity check for odd parity is Correct 0

For P4 - P4 checks locations 4, 5, 6, and 7

There are three 1's in the group

$\therefore$ Parity check for odd parity is Correct ---- 0

For P8 - P8 checks locations 8 and 9

There is one 1's in the group

$\therefore$ Parity check for odd parity is Correct ---- 0 (MSB)

The resultant word is $\neq 0001$

This says that the bit in the number 1 location is in error. It is 1 and should be 0

Therefore, the Correct Code is 101101100.